Module 08 Lab Reflective Journal

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I liked this assignment because it helped me understand why a model can perform well on training data but still make poor predictions on new data. The graphs made the ideas of underfitting, overfitting, and a good fit easier to understand.

I learned that a degree 1 polynomial underfits because it is too simple to follow the pattern in the data. A degree 15 polynomial overfits because it follows the individual data points too closely and creates a jagged line on the graph. The degree 4 polynomial was the best fit because it followed the overall pattern without memorizing the noise.

The learning curves also helped me understand the bias-variance tradeoff. An underfitting model has low training and validation scores because it is too simple. An overfitting model has a large gap between its high training score and lower validation score. This showed me why models need to generalize instead of simply memorizing the training data.